TERMS OF REFERENCE (TOR)

Development of the “GrowTH” Web Application

Child Growth Tracking, Puberty Screening, AI-Assisted Bone Age Assessment

Web Application and Associated Promotional Video Campaign

Prepared for Team: Project Beta

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1. Background and Rationale

Childhood growth and pubertal development are key indicators of a child's overall health. Abnormal patterns — whether growth that is slower or faster than expected, or puberty that begins earlier or later than the typical age range — can signal underlying medical conditions that benefit greatly from early detection. Conditions such as central precocious puberty, if identified late, may result in early closure of the growth plates and a reduced final adult height, in addition to causing psychological and social difficulties for the affected child.

In current clinical practice, screening for these conditions typically requires repeated clinic visits, manual plotting of growth measurements against standard reference charts, and in many cases a bone age assessment performed by reading a hand-and-wrist X-ray. Bone age assessment using traditional methods, such as the Greulich-Pyle (GP) atlas-matching approach or the Tanner-Whitehouse (TW) scoring approach, is time-consuming, requires specialist training, and is subject to variability between different readers. In addition, many healthcare facilities in Thailand do not have ready access to a pediatric endocrinologist or a radiologist experienced in bone age reading, which can delay diagnosis and referral.

Parents, meanwhile, are often the first to notice changes in a child's growth pattern or early signs of pubertal development, but they generally lack a structured, easy-to-use tool to record and interpret these observations between clinic visits. A digital tool that helps parents track growth data over time, screen for early warning signs in a guided manner, and incorporate an automated bone age estimate into that screening process would meaningfully reduce the time between the first parental concern and a clinical diagnosis.

This project responds to that gap by proposing the development of GrowTH, a web application intended for use by parents and caregivers of children, that combines (a) structured growth tracking against standard growth references, (b) a guided puberty development screening questionnaire, and (c) an artificial-intelligence-based bone age prediction feature that reads an uploaded hand X-ray image and returns an estimated bone age to support the screening process. The project also includes a promotional video campaign intended to raise public awareness of puberty disorders, explain the value of early screening, and drive adoption of the application among target users.

This Terms of Reference (TOR) is issued as a class project exercise for third-year students of the Digital Media Engineering (DME) program. It is intended to simulate, as closely as possible, a real-world TOR that a client organization would issue to a vendor or development team, in order to give students hands-on experience with requirements scoping, UX/UI design, full-stack web application development, applied machine learning, and multimedia production within a single, integrated project.

To put the scale of this issue in context, growth and pubertal disorders are not rare conditions. International literature consistently reports that a meaningful minority of children referred to pediatric endocrinology clinics present with concerns related to early or delayed puberty, short stature, or abnormal growth velocity, and that the interval between a parent first noticing a concern and the child receiving a formal clinical evaluation is frequently longer than clinically ideal. Reducing that interval, even modestly, has real value: earlier identification of central precocious puberty, for example, widens the window in which medical treatment, such as hormone-suppressing therapy, can meaningfully protect a child's eventual adult height.

From a technology perspective, this project also serves a pedagogical purpose. It requires the project team to integrate several distinct disciplines — user-centered design, full-stack web engineering, applied machine learning on a real, publicly available medical imaging dataset, and multimedia/video production — into a single coherent product and campaign. This mirrors the kind of cross-functional collaboration expected in professional digital media and software engineering practice, and gives the project team direct, practical experience translating a clinical/health-domain problem into a usable digital product supported by a public-facing communication campaign.

2. Objectives

The objectives of this project are as follows:

1. To design and develop a user-friendly web application (GrowTH) that allows parents to register their children, record growth measurements (height and weight), and view automatically generated growth analysis, including percentile position and standard deviation score (SDS), based on recognized pediatric growth references.
2. To design and develop a structured puberty development screening module within the same application, allowing parents to answer a guided set of questions about physical and developmental signs and receive a preliminary screening result.
3. To design, train, and integrate a machine learning model capable of estimating bone age from an uploaded hand-and-wrist X-ray image, using the RSNA Pediatric Bone Age dataset as the primary training and validation data source, and to make the resulting estimate available within the application as a supporting input to the puberty and growth screening process.
4. To design a complete and consistent user experience and user interface (UX/UI) in Figma covering all in-scope features of the application, prior to development.
5. To build a secure backend system and database capable of storing user accounts, child profiles, growth records, screening responses, and bone age prediction results, designed with appropriate consideration for the sensitivity of children's health data.
6. To produce a coordinated set of promotional video content — comprising a doctor interview, a 2D motion graphic narrative, an application promotional video, an application demonstration video, and a set of short-form video clips — to support the public launch and awareness campaign for GrowTH.
7. To deliver a functioning, demonstrable version of the application and the full set of promotional video assets by the final launch date of November 2.

2A. Assumptions, Target Users, and Constraints

2A.1 Target Users

GrowTH is designed primarily for the following user groups:

- Parents and primary caregivers of children from infancy through adolescence, who wish to track their child's physical growth on an ongoing basis between clinic visits.
- Parents and caregivers who have noticed, or wish to proactively screen for, early signs of pubertal development that may fall outside the typical age range for their child's sex.
- Secondary audience for the promotional video campaign: the broader public, including extended family members and educators, who may benefit from increased awareness of puberty development disorders and may share or recommend the application to parents.

2A.2 Key Assumptions

- The project team will have access to a standard set of pediatric growth reference data (e.g., WHO or national growth standards) suitable for calculating percentiles and SDS values; the specific reference dataset to be used shall be confirmed with the Client Representative early in the project.
- The RSNA Pediatric Bone Age dataset is assumed to be available for download and use under its existing public license for the purposes of model training and academic evaluation.
- The doctor interview video will feature a role-played physician character rather than an interview with an actual practicing physician. The Project Team is responsible for casting a presenter for this role and ensuring the role-player is appropriately briefed. The Client Representative will review and approve the interview script/talking points for clinical accuracy prior to filming, but is not responsible for arranging a real physician's availability.

2A.3 Constraints

- This TOR does not allocate a project budget; the project team shall design and build the solution using tools, datasets, and infrastructure that are free, open-source, or otherwise already available to the team and the institution.
- This TOR does not prescribe a detailed phase-by-phase schedule; the only fixed date is the Final Launch Date of November 2, described in Section 10.
- The application is a class project deliverable and is not intended for clinical deployment or commercial release without further regulatory, clinical-validation, and security review beyond the scope defined here.

3. Scope of Work

The scope of work for this project is organized into two major components: (A) product development of the GrowTH web application, comprising UX/UI design, front-end development, backend and database development, and the AI bone age model; and (B) production of a promotional video campaign in support of the application's public launch. Each sub-section below describes the expected activities and outputs in further detail.

3.1 UX/UI Design (Figma)

The project team shall design the complete user experience and visual interface for the GrowTH web application using Figma, prior to the start of front-end development. The design work shall include, at minimum:

- User research synthesis and definition of primary user personas (e.g., parent/caregiver of a young child, parent/caregiver of a pre-teen or teenager).
- Information architecture and sitemap covering all in-scope screens and navigation paths.
- Low-fidelity wireframes for all core user flows: account registration and login, child profile creation and management, growth data entry, growth result and chart visualization, puberty screening questionnaire and result, bone age X-ray upload and result display, and the knowledge/education section.
- High-fidelity, responsive mockups for desktop and mobile-web breakpoints, applying a consistent visual design system (color palette, typography, iconography, spacing, and component library).
- An interactive Figma prototype sufficient to demonstrate the primary user flows for review and usability walkthroughs with the client representative.
- A documented design handoff package (component specifications, redlines, and exported assets) suitable for direct use by the front-end development team.

All design work shall reflect child-friendly and parent-friendly visual language appropriate to a healthcare-adjacent application, while maintaining a professional and trustworthy appearance consistent with a clinical screening tool.

3.2 Web Application Development (Front End)

The project team shall develop the responsive front-end of the GrowTH web application based on the approved Figma design, implementing all functional modules described in Section 4 (Functional Requirements). Front-end development activities include:

- Implementation of all approved screens and user flows as functioning, responsive web pages compatible with current major desktop and mobile browsers.
- Implementation of data visualization components, including growth charts (height-for-age, weight-for-age, BMI-for-age) with the child's current data point plotted against standard percentile curves.
- Implementation of form-based data entry screens with appropriate input validation (e.g., valid date ranges, plausible height/weight values, required fields) and clear, parent-friendly error messaging.
- Implementation of the bone age X-ray upload interface, including file type/size validation, upload progress indication, and clear presentation of the returned AI prediction result alongside an appropriate explanatory note on how to interpret it.
- Integration with the backend API for all data read/write operations described in Section 3.3.
- Basic accessibility considerations (readable font sizes, sufficient color contrast, descriptive form labels) appropriate for a parent-facing health application.

3.3 Backend and Database Development

The project team shall design and implement a backend system and database to support all application functions. This includes:

- Design of a relational (or otherwise justified) database schema covering, at minimum: user accounts, child profiles, growth measurement records, puberty screening responses and results, bone age prediction records (including the uploaded image reference and the returned prediction), and reference tables for standard growth percentile data.
- Development of a secure, documented application programming interface (API) exposing the operations required by the front end, including authentication, profile management, growth data CRUD operations, screening submission and retrieval, and bone age prediction request/response handling.
- Implementation of user authentication and session management, including password handling in line with standard security practice (e.g., hashed and salted password storage).
- Implementation of role-appropriate access control such that a parent account can only access data belonging to the children profiles linked to that account.
- Implementation of server-side logic for growth metric calculations (percentile/SDS calculation against the standard growth reference data adopted for this project) and for compiling the puberty screening result from submitted questionnaire responses.

- Deployment of the backend, database, and front end to a working environment accessible for demonstration, testing, and grading purposes.
- Basic logging and error handling sufficient to support debugging and to demonstrate sound engineering practice during evaluation.

As this is a class project rather than a production clinical system, the project team is not required to pursue formal regulatory certification (e.g., as a medical device); however, the system shall be designed with the data protection and privacy principles described in Section 6 clearly considered and documented.

3.4 AI Bone Age Prediction Model

The project team shall design, train, and integrate a machine learning model that takes a hand-and-wrist X-ray image as input and returns a predicted bone age as output. This work shall include:

- Acquisition and preparation of the publicly available RSNA Pediatric Bone Age dataset, comprising approximately 12,611 training images and 1,425 validation images, each labeled with bone age (in months) and sex.
- Data preprocessing and augmentation (e.g., resizing, normalization, and techniques such as rotation, flipping, or contrast adjustment as appropriate) to improve model robustness and reduce overfitting.
- Selection and training of an appropriate deep learning architecture (e.g., a convolutional neural network, optionally leveraging transfer learning from a pretrained image classification backbone) for bone age regression, incorporating sex as an auxiliary input where appropriate.
- Model evaluation using a held-out validation/test split, with Mean Absolute Error (MAE), expressed in months, reported as the primary performance metric. The project team shall document the achieved MAE and discuss it against the published results of comparable approaches from the RSNA Pediatric Bone Age Machine Learning Challenge literature.
- Packaging of the trained model and creation of an inference service/endpoint that the backend (Section 3.3) can call to obtain a bone age prediction for a newly uploaded image.
- Documentation of model limitations, including an explicit statement, to be reflected in the application's user interface, that the bone age prediction is a screening aid and does not constitute a clinical diagnosis.

The project team is encouraged, but not required, to explore the use of any additional reference materials (e.g., descriptions of the Greulich-Pyle or Tanner-Whitehouse methods) to inform feature engineering or model interpretation, provided that no copyrighted atlas images are reproduced in any deliverable or in the application itself.

3.5 Promotional Video Production

The project team shall plan, produce, and deliver a coordinated set of video content to support the public awareness and launch campaign for GrowTH. The campaign shall consist of the following five categories of video deliverables:

3.5.1 Doctor Interview Video

A video featuring an interview with a physician (to be arranged with support from the client representative; see Section 8) discussing puberty development disorders — what they are, why early detection matters, and how a screening tool such as GrowTH can support earlier identification and referral. The project team is responsible for developing the interview question guide, filming, audio recording, editing, color correction, subtitling, and final export.

3.5.2 2D Motion Graphic Narrative Video

A 2D animated motion graphic video that tells a short, emotionally engaging story about a parent and child navigating a puberty development disorder — from the early signs being noticed, through the concern and uncertainty the family experiences, to the value of early screening and timely care. This video should be designed primarily for an audience of parents and should avoid clinical or technical jargon in favor of a narrative, story-driven approach. The project team is responsible for the script, storyboard, character and background design, animation, voice-over or narration (as appropriate), sound design/music, and final export.

3.5.3 Application Promotional Video

A general-purpose promotional video introducing GrowTH to potential users, highlighting its core value proposition (growth tracking, puberty screening, AI-assisted bone age estimation) in an engaging, brand-consistent style suitable for use on the project's launch page, social media channels, and any presentation or pitch context.

3.5.4 Application Demonstration Video

A step-by-step walkthrough video demonstrating how to use GrowTH, covering account creation, adding a child profile, recording growth data, completing the puberty screening questionnaire, uploading an X-ray image for bone age prediction, and interpreting the results shown by the application. This video is intended to reduce onboarding friction for new users and may be screen-recorded with voice-over narration.

3.5.5 Short-Form Video Clips

A set of additional short-form video clips (vertical or square format, suitable for social media platforms) designed to attract the attention of the target audience and drive traffic toward the application and the longer-form videos above. The exact number and specific content angles of these clips shall be proposed by the project team and agreed with the client representative during the planning phase described in Section 3.5.6, but should reasonably be expected to include a mix of short educational facts, behind-the-scenes content, and teaser cuts of the longer videos.

3.5.6 Production Process Requirements

For all video deliverables under Section 3.5, the project team shall follow a structured production process consisting of, at minimum: (a) a written script and/or storyboard submitted for client review prior to filming or animation production; (b) a rough-cut or draft version submitted for client feedback; and (c) a final, polished version incorporating client feedback and approved prior to public release. This process is described further under Section 9 (Reporting and Review Process).

4. Functional Requirements

This section describes the functional requirements of the GrowTH web application in further detail, organized by module. A consolidated summary table is also provided in Annex A.

4.1 Account and Profile Management

- FR-1: The system shall allow a new user (parent/caregiver) to register an account using, at minimum, full name, email address, password, and phone number.
- FR-2: The system shall require the user to accept the application's terms of use and privacy notice before account creation is completed.
- FR-3: The system shall allow a registered user to log in and log out securely.
- FR-4: The system shall allow a user to add one or more child profiles, capturing at minimum: child's name, sex, date of birth, and relationship to the user.
- FR-5: The system shall allow a user to view, edit, and switch between multiple child profiles linked to their account.

Rationale: Many households include more than one child, and a parent should be able to manage growth and screening records for each child independently within a single account, without needing to register separately for each child.

4.2 Growth Tracking and Screening

- FR-6: The system shall allow a user to record a growth entry for a selected child, including date of measurement, height, and weight.
- FR-7: The system shall calculate and display the child's percentile and standard deviation score (SDS) for height-for-age and weight-for-age based on an agreed standard growth reference.
- FR-8: For children aged five years and above, the system shall calculate and display BMI and BMI-for-age SDS, with an accompanying plain-language nutritional status summary (e.g., “within normal range”).
- FR-9: The system shall display growth trends over time using line charts for height-for-age, weight-for-age, and BMI-for-age, plotted against standard percentile curves.
- FR-10: The system shall provide a plain-language summary and basic guidance message following each growth entry (e.g., flagging when a result falls notably outside the expected range and recommending consultation with a physician), without presenting this guidance as a clinical diagnosis.
- FR-11: The system shall allow the user to view a complete history of past growth entries for a given child.

Rationale: Growth assessment is inherently longitudinal — a single measurement is far less informative than a trend over time. The historical view and trend charts are therefore considered core, not optional, functionality.

4.3 Puberty Development Screening

- FR-12: The system shall provide a structured, guided questionnaire allowing a parent to report observed signs of pubertal development appropriate to the child's sex and age.
- FR-13: The system shall compile the questionnaire responses into a preliminary screening result, clearly labeled as a screening aid rather than a diagnosis, with guidance on recommended next steps (e.g., consulting a pediatrician or pediatric endocrinologist).
- FR-14: The system shall allow the user to view a history of previously completed puberty screening assessments for a given child.

Rationale: A single screening response is a snapshot; the ability to see how responses change over successive assessments helps a parent and, eventually, a treating physician understand the trajectory of the child's development rather than a single point in time.

4.4 AI-Assisted Bone Age Prediction

- FR-15: The system shall allow a user to upload a hand-and-wrist X-ray image file (in a defined set of accepted formats) for a selected child.
- FR-16: The system shall validate the uploaded file for format and file size, and shall provide clear feedback if the upload does not meet requirements.
- FR-17: The system shall submit the uploaded image to the bone age prediction model (Section 3.4) and display the predicted bone age result to the user within a reasonable processing time.
- FR-18: The system shall present the bone age result together with an explanatory note describing what the figure means, its margin of error, and that it is intended to support — not replace — clinical assessment.
- FR-19: The system shall associate each bone age prediction result with the corresponding child profile and growth/screening history, allowing the parent to review past predictions over time.

Rationale: The bone age feature is most clinically meaningful when considered alongside the child's growth and puberty screening history rather than in isolation; linking these data points within the application supports a more complete picture for both the parent and any physician the parent later consults.

4.5 Knowledge / Education Section

- FR-20: The system shall provide a knowledge section containing reference articles and/or reference growth chart materials relevant to child growth and puberty development, sourced or adapted with appropriate attribution and without infringing copyright.

- FR-21: Content in the knowledge section shall be organized and searchable/browsable by topic.

4.6 General / Non-Functional Considerations Relevant to Functionality

- FR-22: The system shall be usable on both desktop and mobile web browsers through a responsive layout.
- FR-23: The system shall display all clinical or quasi-clinical results (growth status, screening result, bone age prediction) using plain, parent-friendly language, supplemented with the underlying figures for users who want more detail.
- FR-24: The system shall ensure that a given user can only access the profiles and data of children linked to their own account.

5. Deliverables

The following deliverables shall be submitted by the project team over the course of the project. A consolidated checklist version of this table is provided in Annex B for use during milestone review meetings.

No.	Deliverable	Description / Format	Related Section
D1	UX/UI Design Package	Complete Figma project file (with editable components), exported high-fidelity screens (PDF/PNG), and an interactive prototype link	3.1
D2	Web Application (Front End)	Deployed, responsive web application implementing all approved screens and flows; complete front-end source code repository	3.2
D3	Backend, API, and Database	Deployed backend service; documented API specification; database schema documentation; backend source code repository	3.3
D4	AI Bone Age Prediction Model	Trained model file/artifact; model training and evaluation report (including MAE results); inference service integrated with the backend	3.4
D5	Doctor Interview Video	Final edited video file (with subtitles), including the interview question guide used	3.5.1
D6	2D Motion Graphic Narrative Video	Final edited animated video file, including script and storyboard	3.5.2
D7	Application Promotional Video	Final edited promotional video file	3.5.3
D8	Application Demonstration Video	Final edited how-to-use video file	3.5.4
D9	Short-Form Video Clips	A set of short-form clips (quantity and angles as agreed during planning), delivered in formats suitable for social media platforms	3.5.5
D10	Project Documentation Set	System overview document, user manual / quick guide for parents, and a final project report summarizing scope delivered, known limitations, and recommendations for future work	All
D11	Source Files and Handover Package	All raw and editable production files (Figma, video project files, model training code/notebooks, application source code) handed over to the client representative	All

All deliverables shall be submitted in a format that allows the client representative to independently open, review, and (where applicable) edit the material without requiring proprietary software licenses not already available to the client, except where industry-standard professional tools (e.g., Figma, video editing software) are clearly necessary and commonly available.

6. Technical Requirements and Standards

6.1 Platform and Compatibility

- The application shall be delivered as a responsive web application, accessible via current versions of major web browsers (e.g., Chrome, Safari, Edge, Firefox) on both desktop and mobile devices.
- The application shall be hosted in an environment accessible via a stable URL for the duration of the evaluation and demonstration period.
- The project team shall select and justify their technology stack (front-end framework, backend framework/language, database system, and model-serving approach) in the system overview document (Deliverable D10), considering maintainability, suitability for the dataset and model type, and feasibility within the project timeline.

6.2 Data Privacy and Protection

Given that GrowTH collects and processes health-related information about children, which is sensitive personal data, the project team shall design the system with the following principles in mind, consistent with the spirit of Thailand's Personal Data Protection Act (PDPA):

- Collection of personal data shall be limited to what is necessary for the application's stated functions (data minimization).
- The application shall present a clear privacy notice explaining what data is collected, how it is used, and how it is stored, prior to account registration.
- Uploaded X-ray images and associated prediction results shall be stored securely and shall be accessible only to the account that uploaded them.
- Passwords shall never be stored in plain text; an industry-standard hashing approach shall be used.
- The project team shall document, in the system overview document, what data is collected, where and how it is stored, and what (if any) data would need to be deleted upon a user's request, to demonstrate awareness of data subject rights even though full PDPA compliance certification is not required for this class project.

6.3 AI Model Standards

- The bone age prediction model shall be trained, validated, and tested using a clearly documented data split, with no overlap between training and test data.
- Model performance shall be reported using Mean Absolute Error (MAE) in months on the held-out test set, alongside any other metrics the project team considers informative (e.g., Mean Squared Error).

- The project team shall document any data augmentation, preprocessing, or class-balancing techniques applied, and shall discuss any observed limitations of the model (e.g., performance variation by age range or sex).
- The application shall clearly present the bone age prediction as a screening aid, not a diagnostic result, in the user interface itself (not only in supporting documentation).

6.4 Video Production Standards

Attribute	Requirement
Resolution	Minimum Full HD (1920×1080) for long-form videos (doctor interview, motion graphic, promo, demo); short-form clips delivered in a vertical or square format suitable for social platforms
Audio	Clear, intelligible audio for all spoken content; background music, where used, shall not overpower narration or dialogue
Language and Subtitles	Primary narration/dialogue in Thai, with English subtitles (or vice versa, as agreed with the client representative) for all long-form videos
Branding	Consistent use of the GrowTH name, logo, and visual identity (derived from the approved Figma design system) across all video deliverables
File Format	Delivered in a standard, widely compatible video container/codec (e.g., MP4/H.264), with editable project files also provided as part of the handover package
Content Sensitivity	All content involving discussion of puberty development and the featured parent-child narrative shall be handled with appropriate sensitivity, avoiding stigmatizing language and avoiding identification of any real child or patient without explicit, informed consent

6.5 Quality and Testing

- The project team shall perform functional testing of all features listed in Section 4 prior to each milestone review, and shall maintain a basic test record or checklist as part of the project documentation.
- The project team shall test the application across at least two different browsers and at least one mobile device or mobile viewport size prior to final delivery.
- The project team shall address defects identified during client review within a reasonable timeframe agreed during the relevant milestone meeting, consistent with the overall schedule culminating in the November 2 launch date.

7. Project Team Qualifications

As this project is conducted as a class assignment for third-year students of the Digital Media Engineering program, qualifications are framed around relevant coursework, demonstrated skills, and team composition rather than professional work history. The project team is expected to be a multidisciplinary group of students collectively able to cover the following skill areas:

- UX/UI Design: Working proficiency in Figma, including component-based design systems, responsive layout principles, and prototyping; familiarity with basic usability evaluation methods.
- Front-End Development: Working knowledge of a modern web front-end framework (e.g., React, Vue, or equivalent), responsive CSS techniques, and consumption of REST or equivalent APIs.
- Backend and Database Development: Working knowledge of a backend framework and language (e.g., Node.js, Python/Flask or Django, or equivalent), relational database design, and basic API security practices (authentication, input validation).
- Applied Machine Learning: Coursework or project experience in machine learning or deep learning, including practical experience with an image-based deep learning framework (e.g., PyTorch or TensorFlow/Keras), and a basic understanding of model evaluation methodology.
- Video Production and Motion Graphics: Coursework or project experience in video production (filming, interview-style shooting, and editing) and 2D motion graphic / animation production, including storyboard development and narrative scriptwriting.
- Project Coordination: At least one member of the team shall act as Project Manager, responsible for coordinating internal task assignment, tracking progress against milestones, and serving as the primary point of contact with the client representative.

It is expected that individual team members may hold overlapping or combined responsibilities across the above areas, consistent with the size of a typical class project team. The course instructor(s) may, at their discretion, advise on team formation to ensure reasonable coverage of the skill areas listed above.

8. Roles and Responsibilities

This section sets out the respective responsibilities of the Client Representative and the Project Team for the duration of the project.

8.1 Client Representative Responsibilities

- Act as the primary decision-maker and approver for scope, design, and content decisions throughout the project.
- Attend and participate in the weekly project update meeting described in Section 9.
- Review and provide timely feedback on submitted drafts (designs, application builds, scripts, storyboards, and video cuts) within an agreed turnaround time so as not to delay the project schedule.
- Arrange access to a physician for the purpose of the doctor interview video (Section 3.5.1), including coordinating the physician's availability and obtaining any necessary consent for filming.
- Provide or review clinical/technical content used in the application (e.g., reference growth standards, screening questionnaire content, explanatory text accompanying the bone age result) to confirm accuracy prior to implementation.
- Provide reasonable access to any reference materials, prior project documentation, or institutional branding guidelines relevant to the project.
- Confirm final acceptance of each deliverable in accordance with Section 12.

8.2 Project Team Responsibilities

- Deliver all items described in Sections 3, 4, and 5 in accordance with the technical requirements set out in Section 6.
- Assign a Project Manager who will coordinate the team's work, maintain a task/progress tracker, and represent the team in weekly meetings with the Client Representative.
- Proactively flag risks, blockers, or scope questions to the Client Representative as early as possible rather than at the point of a missed milestone.
- Prepare and present progress updates, drafts, and demonstrations at each weekly meeting and at each milestone review described in Section 9.
- Incorporate reasonable feedback from the Client Representative within the agreed revision process for each deliverable type.
- Maintain version control and basic documentation practices (e.g., a shared repository for code, a shared drive or Figma workspace for design and video assets) throughout the project.

- Prepare all materials necessary for the final acceptance review described in Section 12.

8.3 Shared / Collaborative Responsibilities

- Both parties shall jointly agree on the script and content angle for each video deliverable prior to production.
- Both parties shall jointly review and agree on the final list and content angles for the short-form video clips described in Section 3.5.5 during the planning phase.
- Both parties shall jointly maintain a shared issue/feedback log to track open items, their status, and resolution.

9. Reporting and Review Process

To maintain alignment between the Project Team and the Client Representative and to ensure the project remains on track toward the November 2, launch date, the following reporting and review process shall be followed throughout the project.

9.1 Weekly Update Meeting

The Client Representative and the Project Manager shall meet once per week for the duration of the project. The purpose of this weekly meeting is to:

- Review progress made since the previous meeting against the current task/progress tracker.
- Surface and resolve any open questions, blockers, or scope clarifications.
- Preview any in-progress design, development, or video work, where available, even if not yet at a formal milestone.
- Agree on priorities and expected progress for the upcoming week.

The Project Manager is responsible for preparing a short written status update (e.g., a one-page summary or shared tracker snapshot) ahead of each weekly meeting, and for circulating brief meeting notes afterward summarizing decisions and action items.

9.2 Milestone Reviews

In addition to the weekly update meeting, the following milestone reviews shall be scheduled at the relevant points in the project, each requiring formal sign-off from the Client Representative before the team proceeds to the next phase of work:

Milestone	Review Focus
M1 – UX/UI Design Review	Review of wireframes and high-fidelity Figma designs and prototype against the requirements in Sections 3.1 and 4
M2 – Application Beta 1 Review	Review of a working but incomplete build of the front end and backend, demonstrating core data flows (account creation, growth entry, screening flow)
M3 – AI Model Review	Review of the trained bone age model's evaluation results (including MAE) and a demonstration of the integrated prediction flow within the application
M4 – Video Script/Storyboard Review	Review and approval of scripts and/or storyboards for all five video categories described in Section 3.5, prior to filming/animation production
M5 – Application Beta 2 Review	Review of a feature-complete application build, including all functional requirements in Section 4, for user acceptance testing
M6 – Video Rough-Cut Review	Review of rough-cut versions of all video deliverables for feedback prior to final editing

M7 – Final Acceptance Review	Final review of all deliverables described in Section 5 against the acceptance criteria in Section 12, ahead of the November 2 launch date

Where a milestone review identifies required changes, the Project Team shall propose a revised completion date for the affected items, and this shall be discussed and agreed at the next weekly update meeting.

9.3 Revision Rounds

For design and video deliverables in particular, the following standard revision allowance shall apply unless otherwise agreed: up to two rounds of consolidated feedback and revision per deliverable (e.g., two rounds of feedback on a Figma design, two rounds of feedback on a video rough cut). Additional revision rounds beyond this allowance may be accommodated by agreement between both parties, with appropriate adjustment to the affected portion of the schedule.

10. Project Timeline Anchor

This TOR does not prescribe a detailed phase-by-phase schedule or budget. The Project Team is responsible for proposing and maintaining its own internal working schedule and task allocation, to be shared with the Client Representative in the form of a task/progress tracker as referenced in Section 9.1.

The single fixed schedule anchor for this project is the Final Launch Date of November 2, by which date all deliverables listed in Section 5 shall be complete, reviewed, and accepted in accordance with Section 12, and the GrowTH application and full promotional video campaign shall be ready for public release.

The Project Team is strongly advised to work backward from this date when planning internal milestones, allowing adequate buffer time for the revision rounds described in Section 9.3 and for the final acceptance review (Milestone M7).

11. Intellectual Property and Confidentiality

11.1 Intellectual Property

- All deliverables produced under this project — including but not limited to Figma design files, application source code, the trained AI model and associated training code, video footage, animation assets, and project documentation — shall be considered the property of the academic project (Department of Computer Engineering / Digital Media Engineering Program) upon final acceptance, for use in course evaluation, academic portfolio purposes, and any future continuation of the project, subject to applicable university policy on student work.
- The Project Team retains the right to reference and showcase the project (e.g., in a personal portfolio or resume) unless otherwise restricted by university policy, provided that any sensitive data (e.g., real patient X-ray images, if used) is not included in such showcases.
- Any third-party assets used in the project (e.g., stock music, fonts, icon libraries, or the RSNA Pediatric Bone Age dataset) shall be properly licensed or used in accordance with their applicable terms of use, and shall be documented in a credits/attribution list included in the final project documentation (Deliverable D10).
- No copyrighted reference materials (e.g., pages from the Greulich-Pyle or Tanner-Whitehouse atlases) shall be reproduced within the application, the documentation, or any video deliverable.

11.2 Confidentiality and Data Handling

- Any real patient data, real X-ray images, or identifiable clinical information shared with the Project Team for reference or testing purposes (if any) shall be treated as confidential, used only for the purposes of this project, and shall not be redistributed, published, or included in any public-facing deliverable, including the promotional videos.
- Where the application is demonstrated or tested using sample data, the Project Team shall use synthetic, anonymized, or publicly available dataset images (e.g., from the RSNA Pediatric Bone Age dataset) rather than real, identifiable patient data, unless explicit written consent has been obtained and confirmed by the Client Representative.
- Any individual appearing in the video deliverables (e.g., the interviewed physician, or any actor/model used to represent a parent or child in the motion graphic or demonstration video) shall provide informed consent for filming and for the intended public use of the resulting footage.
- The Project Team shall not disclose details of the project, including unreleased designs, footage, or model performance results, to parties outside

the course context without the Client Representative's agreement, prior to the public launch.

12. Acceptance Criteria and Evaluation

Final acceptance of the project shall be based on the Final Acceptance Review (Milestone M7, Section 9.2), at which the Client Representative will assess each deliverable against the criteria below. Acceptance criteria are intended both to confirm completion of the TOR's scope and to serve as a basis for course evaluation of the student project team.

12.1 Application Acceptance Criteria

- All functional requirements listed in Section 4 are implemented and demonstrable in the deployed application.
- The application reflects the approved Figma design (Deliverable D1) in terms of layout, visual identity, and core user flows, with any deviations explained and agreed.
- The bone age prediction model is integrated end-to-end (image upload through to displayed result) and its documented MAE performance is consistent with the evaluation report submitted at Milestone M3.
- The application functions correctly across the browsers and device sizes specified in Section 6.5, with no critical defects (i.e., defects that prevent completion of a core user flow) outstanding at the time of review.
- Data privacy considerations described in Section 6.2 are demonstrably reflected in the application (e.g., a visible privacy notice, secure password handling, account-scoped data access).

12.2 Video Campaign Acceptance Criteria

- All five video categories described in Section 3.5 are delivered in final form, meeting the technical standards in Section 6.4.
- Each video reflects the script/storyboard approved at Milestone M4, with any agreed changes from the rough-cut review (Milestone M6) incorporated.
- Content involving discussion of puberty development disorders is accurate, appropriately sensitive, and consistent with the information reviewed by the Client Representative.
- All necessary consents (physician, any individuals appearing on camera) have been obtained and documented.

12.3 Documentation Acceptance Criteria

- The full documentation set (Deliverable D10) and source/handover package (Deliverable D11) are complete, organized, and delivered in a form usable without the continued involvement of the Project Team.
- The AI model evaluation report clearly documents methodology, results (including MAE), and known limitations.

12.4 Overall Acceptance

The project shall be considered formally accepted when the Client Representative confirms, in writing (e.g., via a signed acceptance checklist based on Annex B), that all deliverables meet the criteria above, or that any outstanding items have been explicitly waived or rescheduled by mutual agreement. Formal acceptance is targeted to occur on or before the Final Launch Date of November 2.

13. Risk Management

The following risks have been identified as reasonably foreseeable for this project. The Project Team shall monitor these risks throughout the project and raise any materializing issues at the weekly update meeting described in Section 9.1, together with a proposed mitigation or schedule adjustment.

Risk	Potential Impact	Suggested Mitigation
AI model does not reach an acceptable MAE within the available training time	Reduced confidence in the bone age feature; possible need to clearly label the feature as experimental	Begin model training early; use transfer learning from a pretrained backbone; document achieved performance transparently rather than overstating accuracy
Physician availability for the interview video is delayed	Doctor Interview Video (D5) and downstream editing schedule slip	Client Representative to begin coordinating physician availability as early as possible in the project; project team to prepare interview questions in advance so filming can proceed quickly once scheduled
Scope creep from features outside the requirements	Reduced time available for in-scope deliverables	Reconfirm scope boundaries (Section 2, exclusions) at each milestone review; any proposed scope changes to be raised explicitly with the Client Representative rather than implemented ad hoc
Limited free/student-tier infrastructure capacity (hosting, storage, or model inference compute)	Performance issues or downtime during demonstration or evaluation	Select lightweight, efficient model architectures where feasible; test deployment well ahead of milestone reviews and the final launch date
Sensitive subject matter (puberty development) handled without sufficient care in video content	Risk of content being perceived as insensitive, stigmatizing, or clinically inaccurate	Clinical content to be reviewed by the Client Representative prior to script approval (Milestone M4); maintain a respectful, factual, non-alarmist tone throughout
Team member availability conflicts with other coursework or commitments	Delays to internal task completion	Project Manager to maintain a realistic internal schedule with buffer time, working backward from the November 2 launch date, and to redistribute tasks as needed

Annex A: Functional Requirement Summary Table

ID	Requirement Summary	Module
FR-1 to FR-5	Account registration, login, terms acceptance, child profile management	Account & Profile
FR-6 to FR-11	Growth entry, percentile/SDS calculation, BMI-for-age, trend charts, guidance messages, history view	Growth Tracking
FR-12 to FR-14	Guided puberty screening questionnaire, preliminary result, history view	Puberty Screening
FR-15 to FR-19	X-ray upload, validation, AI bone age prediction, result explanation, history view	AI Bone Age
FR-20 to FR-21	Knowledge section content and organization	Knowledge / Education
FR-22 to FR-24	Responsive design, plain-language results, account-scoped data access	General / Non-Functional

Annex B: Deliverable Checklist

No.	Deliverable	Submitted (Y/N)	Accepted (Y/N)	Notes
D1	UX/UI Design Package			
D2	Web Application (Front End)			
D3	Backend, API, and Database			
D4	AI Bone Age Prediction Model			
D5	Doctor Interview Video			
D6	2D Motion Graphic Narrative Video			
D7	Application Promotional Video			
D8	Application Demonstration Video			
D9	Short-Form Video Clips			
D10	Project Documentation Set			
D11	Source Files and Handover Package			

Annex C: Glossary of Terms

Term	Definition
Bone Age	An estimate of skeletal maturity derived from the appearance of bones (typically of the hand and wrist) on an X-ray, compared against population reference standards.
Precocious Puberty	The onset of pubertal physical changes at an earlier age than typical, which can be associated with accelerated bone maturation and reduced final adult height if untreated.
MAE (Mean Absolute Error)	A model evaluation metric representing the average absolute difference between predicted and actual values; in this project, expressed in months of bone age.
Percentile / SDS	Statistical measures used to express how a child's height, weight, or BMI compares to a reference population of children of the same age and sex; SDS stands for Standard Deviation Score (also known as a z-score).
RSNA Pediatric Bone Age Dataset	A publicly available dataset of labeled hand-and-wrist X-ray images, compiled by the Radiological Society of North America, widely used for training and benchmarking automated bone age prediction models.
PDPA	Thailand's Personal Data Protection Act, which sets requirements for the lawful collection, use, and protection of personal data, including health-related data.
Figma	A cloud-based collaborative interface design tool used for creating wireframes, high-fidelity mockups, and interactive prototypes.
UX/UI	User Experience and User Interface design — respectively, the overall usability and flow of an application, and its visual presentation and interactive elements.

Annex D: Indicative Puberty Screening Question Categories

The following question categories are provided as an indicative starting point for the puberty development screening questionnaire (FR-12). The exact wording, clinical thresholds, and scoring logic shall be finalized in consultation with the Client Representative and, where appropriate, reviewed against established clinical screening references before implementation.

D.1 For Female Children

- Has breast development (thelarche) been observed, and at approximately what age?
- Has pubic or underarm hair growth been observed, and at approximately what age?
- Has menstruation begun, and at approximately what age?
- Has the child experienced a noticeable recent increase in height growth rate?

D.2 For Male Children

- Has testicular or genital enlargement been observed, and at approximately what age?
- Has pubic, underarm, or facial hair growth been observed, and at approximately what age?
- Has voice deepening been observed?
- Has the child experienced a noticeable recent increase in height growth rate?

D.3 General Questions (Both Sexes)

- Family history: at approximately what age did the child's parents or siblings begin puberty?
- Has the child shown any behavioral, mood, or skin changes (e.g., acne, body odor) that the parent associates with early physical development?
- Are there any other health conditions, medications, or relevant medical history the parent wishes to record?

The result of this questionnaire shall be compiled into a plain-language screening summary, as described under FR-13, and shall not be presented as a clinical diagnosis.